



Chapter 2

Logic Algebra II





2.3 Forms and Transformations of Logical Function Expressions

In any logic function, the form of its expression is not unique. The following describes the **Common form** and the **Canonical form** of the logic function expression; and their conversion.

2.3.1 Two common forms

Two common forms: “AND-OR” and “OR-AND” forms.

I. “AND-OR” FORM

“AND-OR”FORM: is an “OR” expression with many “AND” terms ; e.g.,

$$F = \overline{A}B + A\overline{B}C + \overline{C}$$

“AND” term is also called “PRODUCT” term, correspondingly, “AND—OR” form is called “SUM OF PRODUCT” form.





II. “OR-AND” form

“OR-AND” form : is an **“AND”** expression with many **“OR”** terms ; e.g.,

$$F(A, B, C, D) = (\bar{A} + B)(B + \bar{C})(A + \bar{B} + C)D$$

“OR” term is also called **“SUM”** term, correspondingly, **“OR—AND”** form is called **“PRODUCT OF SUM”** form.





$$F(A, B, C) = (A\bar{B} + C)(A + B\bar{C}) + B$$

neither a “AND—OR” form, nor “OR—AND” form !

The common forms of logic functions are not unique. E.g

$$F = AB + \bar{A}C + BC = AB + \bar{A}C$$

2.3.2 Canonical form

In order to make logical functions correspond to unique logical expressions in the study of logical problems, the **canonical** form of logical function expressions was introduced. The canonical form of logical function expressions is based on the concepts of the **minterms** and **maxterms**.





I. Minterm and Maxterm

1. Minterm

(1) Definition: Minterms are made of AND terms with every variable present in either true or complemented form. It is also called canonical AND term

(2) Amount: n variables form 2^n minterms

e.g., 3 variables A, B, C can form $\bar{A} \cdot \bar{B} \cdot \bar{C}$, $\bar{A} \cdot \bar{B} \cdot C$, ..., $A B C$, 8 minterms in total.

(3) Abbreviated: m_i for the i th minerm.

i: The index for the minterm or maxterm, expressed as a binary number, is used to determine whether the variable is shown in the true form or complemented form.





- **For Minterms:**

- “1” means the variable is “Not Complemented” (true) and
- “0” means the variable is “Complemented”.

e.g., 3 variable A,B,C have a minterm $\bar{A} \bar{B} C$, which is represented by m_5 , since

$$\begin{array}{ccc} A & \bar{B} & C \longrightarrow m_5 \\ \downarrow & \downarrow & \downarrow \\ 1 & 0 & 1 \longrightarrow (5)_{10} \end{array}$$





A	B	C	Minterm	Index
0	0	0	$\overline{A} \overline{B} \overline{C}$	m_0
0	0	1	$\overline{A} \overline{B} C$	m_1
0	1	0	$\overline{A} B \overline{C}$	m_2
0	1	1	$\overline{A} B C$	m_3
1	0	0	$A \overline{B} \overline{C}$	m_4
1	0	1	$A \overline{B} C$	m_5
1	1	0	$A B \overline{C}$	m_6
1	1	1	$A B C$	m_7





(4) Properties, the minterm has four properties:

Property 1: Each minterm has one and only one combination representing 1 in the 2^n terms (a minimum of 1s). All other entries are 0.

$x \ y$	m_0	m_1	m_2	m_3
0 0	1	0	0	0
0 1	0	1	0	0
1 0	0	0	1	0
1 1	0	0	0	1





Property 2: With the same variables, the “AND” result of two different minterms **is 0**.

Since impossible to find two minterms whose values are 1 at the same time, thus:

$$m_i \cdot m_j = 0$$

Property 3: **the sum of** all minterms of n variables is 1. Usually use the accumulation of symbols in Mathematics “ Σ ”, thus:

$$\sum_{i=0}^{2^n - 1} m_i = 1$$

Property 4: Minterm made up of N variables has N adjacent minterms.

Adjacent minterms : For two minterms, **one variable is complemented, but the rest are identical**. E.g.,, A three-variable minterm $A B C$, its adjacent minterm includes $\overline{A} B C$





2. Maxterm

Definition : Maxterms are OR terms with every variable present in either true or complemented form. It is also called canonical OR term.

Amount: n variables form 2^n maxterms

e.g., 3 variables A, B, C can form $A+B+C, A+B+\bar{C}, \dots, \bar{A}+\bar{B}+\bar{C}$ 8 maxterms in total.





- **Shorthand:** use M_i to present the *i*th maxterm.

The rule of i : The index for the maxterm, expressed as a binary number, is used to determine whether the variable is shown in the true form or complemented form.

For Maxterms:

“0” means the variable is “Not Complemented” (true) and

“1” means the variable is “Complemented”.





e.g., 3 variables A,B,C can form $\overline{A} + B + \overline{C}$, which is represented by M_5 , since

$$\begin{array}{ccc} \overline{A} & + & B & + & \overline{C} & \longrightarrow & M_5 \\ \downarrow & & \downarrow & & \downarrow & & \\ 1 & & 0 & & 1 & \longrightarrow & (5)_{10} \end{array}$$





A	B	C	Maxterm	Index
0	0	0	$A + B + C$	M_0
0	0	1	$A + B + \overline{C}$	M_1
0	1	0	$A + \overline{B} + C$	M_2
0	1	1	$A + \overline{B} + \overline{C}$	M_3
1	0	0	$\overline{A} + B + C$	M_4
1	0	1	$\overline{A} + B + \overline{C}$	M_5
1	1	0	$\overline{A} + \overline{B} + C$	M_6
1	1	1	$\overline{A} + \overline{B} + \overline{C}$	M_7





Properties: the maxterm has four properties

Property 1 Each maxterm has one and only one combination representing 0 in the 2^n terms. All other entries are 1.

For any OR terms, the maxterm has the maximum number of combinations whose values are 1 (a maximum of 1s). That is why it is called the maxterm.

$x y$	M_0	M_1	M_2	M_3
0 0	0	1	1	1
0 1	1	0	1	1
1 0	1	1	0	1
1 1	1	1	1	0





Property 2 : With the same variables, the “OR” result of two different maxterms **is 1**.

Since impossible to find two maxterms whose values are 0 at the same time, thus the OR result is 1.

$$M_i + M_j = 1$$

Property 3 The AND result of all maxterms of n variables is 0. Usually use the accumulate multiplication symbol “ Π ”, thus,

$$\prod_{i=0}^{2^n-1} M_i = 0$$

Property 4: A Maxterm made up of N variables has N adjacent maxterms.

Adjacent maxterms : For two maxterms, one variable is complemented, but the rest are identical.





A truth table of the minterms and maxterms of 2-variable

Variable		Minterm				Maxterm			
A	B	$\bar{A} \cdot \bar{B}$	$\bar{A}B$	$A\bar{B}$	AB	$A + B$	$A + \bar{B}$	$\bar{A} + B$	$\bar{A} + \bar{B}$
		m_0	m_1	m_2	m_3	M_0	M_1	M_2	M_3
0	0	1	0	0	0	0	1	1	1
0	1	0	1	0	0	1	0	1	1
1	0	0	0	1	0	1	1	0	1
1	1	0	0	0	1	1	1	1	0





3. Minterm and Maxterm Relationship

With the same variables, the minterm and maxterm with the identical i , are complemented to each other. Giving m_i , its complement is M_i . Thus we have:

$$\overline{m_i} = M_i \quad \text{or} \quad m_i = \overline{M_i} \quad \text{why?}$$

E.g., 3 variables: A,B,C form a minterm m_3 and a maxterm M_3 , then, they have:

$$\overline{m_3} = \overline{ABC} = A + \overline{B} + \overline{C} = M_3$$

$$\overline{M_3} = \overline{A + \overline{B} + \overline{C}} = \overline{A}BC = m_3$$





II. Canonical Forms

Any Boolean function can be expressed as a Canonical Sum of Minterms and a Canonical Produce of Maxterm, such two forms.

1. Canonical Sum of Minterms

A logic expression is made of a number of minterm by ORing together, is called Canonical Sum of Minterms, it is also called Minterm expression.

E.g., a 3 variable canonical sum of minterm is defined:

$$F(A,B,C) = \bar{A} \cdot \bar{B}C + \bar{A}B\bar{C} + A\bar{B} \cdot \bar{C} + ABC$$

The expression can be denoted in the formal shorthand

$$\begin{aligned} F(A, B, C) &= m_1 + m_2 + m_4 + m_7 \\ &= \Sigma m(1,2,4,7) \end{aligned}$$





2. Canonical Product of Maxterms

A logic expression is made of a number of maxterm by ANDing together, is called Canonical Product of Maxterms, it is also called Maxterm expression.

E.g. , $A + B + \bar{C}$, $\bar{A} + B + \bar{C}$, $\bar{A} + \bar{B} + \bar{C}$ are three maxterms made of three variables, use the AND process to combine the three terms, then, we have a three-variable Canonical Product of Maxterm:

$$F(A, B, C) = (A + B + \bar{C})(\bar{A} + B + \bar{C})(\bar{A} + \bar{B} + \bar{C})$$

The expression can be denoted in the formal shorthand

$$F(A, B, C) = M_1 M_5 M_7 = \prod M(1, 5, 7)$$





Minterm Function Example

- Example: Find $F_1 = m_1 + m_4 + m_7$
- $F_1 = \bar{x} \bar{y} z + x \bar{y} \bar{z} + x y z$

x y z	index	$m_1 + m_4 + m_7 = F_1$					
0 0 0	0	0	+	0	+	0	= 0
0 0 1	1	1	+	0	+	0	= 1
0 1 0	2	0	+	0	+	0	= 0
0 1 1	3	0	+	0	+	0	= 0
1 0 0	4	0	+	1	+	0	= 1
1 0 1	5	0	+	0	+	0	= 0
1 1 0	6	0	+	0	+	0	= 0
1 1 1	7	0	+	0	+	1	= 1





Minterm Function Example

- $F(A, B, C, D, E) = m_2 + m_9 + m_{17} + m_{23}$
- $F(A, B, C, D, E) =$





Maxterm Function Example

- Example: Implement F1 in maxterms:

$$F_1 = M_0 \cdot M_2 \cdot M_3 \cdot M_5 \cdot M_6$$

$$F_1 = (x + y + z) \cdot (x + \bar{y} + z) \cdot (x + \bar{y} + \bar{z}) \\ \cdot (\bar{x} + y + \bar{z}) \cdot (\bar{x} + \bar{y} + z)$$

x y z	i	$M_0 \cdot M_2 \cdot M_3 \cdot M_5 \cdot M_6 = F1$
0 0 0	0	$0 \cdot 1 \cdot 1 \cdot 1 \cdot 1 = 0$
0 0 1	1	$1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 = 1$
0 1 0	2	$1 \cdot 0 \cdot 1 \cdot 1 \cdot 1 = 0$
0 1 1	3	$1 \cdot 1 \cdot 0 \cdot 1 \cdot 1 = 0$
1 0 0	4	$1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 = 1$
1 0 1	5	$1 \cdot 1 \cdot 1 \cdot 0 \cdot 1 = 0$
1 1 0	6	$1 \cdot 1 \cdot 1 \cdot 1 \cdot 0 = 0$
1 1 1	7	$1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 = 1$





Maxterm Function Example

- $F(A, B, C, D) = M_3 \cdot M_8 \cdot M_{11} \cdot M_{14}$
- $F(A, B, C, D) =$





2.3.3 Conversion to canonical forms

There are two common ways to convert a logical function expression into the canonical form.

I. Algebraic conversion

The algebraic conversion method uses logical properties, theorems, and rules to perform logical transformations, transforming function expressions from one form to another.

1 . To obtain canonical SOM

Common steps:

Step 1: Convert an expression to common “AND-OR” ;

Step 2: For the common expressions, expand all terms first to have all variables. Do this by “ANDing” any term missing a variable x with a term

$$X = X(Y + \bar{Y})$$





E.g. Convert the algebra function $F(A,B,C) = \overline{(\overline{A\overline{B}} + \overline{B\overline{C}})} \cdot \overline{A\overline{B}}$

To SOM

A: Step 1: Convert to common “AND-OR”, i.e.,

$$\begin{aligned} F(A,B,C) &= \overline{(\overline{A\overline{B}} + \overline{B\overline{C}})} \cdot \overline{A\overline{B}} \\ &= \overline{A\overline{B}} + \overline{B\overline{C}} + A\overline{B} \\ &= (\overline{A} + B)(\overline{B} + C) + A\overline{B} \\ &= \overline{A} \cdot \overline{B} + \overline{A}C + BC + A\overline{B} \end{aligned}$$

Step2: Find the missing variables in the common “AND-OR”, expand the terms to SOM :

$$\begin{aligned} F(A,B,C) &= \overline{A} \cdot \overline{B}(\overline{C} + C) + \overline{A}C(\overline{B} + B) + (\overline{A} + A)BC + A\overline{B}(\overline{C} + C) \\ &= \overline{A} \cdot \overline{B} \cdot \overline{C} + \overline{A} \cdot \overline{B} \cdot C + \overline{A} \cdot \overline{B} \cdot C + \overline{A}BC + \overline{A}BC + ABC + A\overline{B}\overline{C} + ABC \\ &= \overline{A} \cdot \overline{B} \cdot \overline{C} + \overline{A} \cdot \overline{B} \cdot C + \overline{A}BC + A\overline{B}\overline{C} + ABC \end{aligned}$$





we can use the shorthand:

$$\begin{aligned} F(A,B,C) &= m_0 + m_1 + m_3 + m_6 + m_7 \\ &= \sum m(0,1,3,6,7) \end{aligned}$$

Somehow, the algebra expression is of common “AND-OR”, we can proceed the second step directly.

2 . To obtain canonical POM

Common steps:

Step 1: Convert an expression to common “OR-AND” ;

Step 2: expand all terms first to contain all variables. Do this by applying $A = (A + B)(A + \bar{B})$ again and again.





E.g, convert the logic function $F(A,B,C) = \overline{(AB + \overline{A}C)} + \overline{B}C$ **to**
POM:

A: Step 1: Convert to common “OR-AND” , i.e,

$$\begin{aligned} F(A,B,C) &= \overline{(AB + \overline{A}C)} + \overline{B}C \\ &= \overline{AB} \cdot \overline{\overline{A}C} + \overline{B}C \\ &= (\overline{A} + \overline{B})(A + \overline{C}) + \overline{B}C \\ &= [(\overline{A} + \overline{B})(A + \overline{C}) + \overline{B}][(\overline{A} + \overline{B})(A + \overline{C}) + C] \\ &= (\overline{A} + \overline{B} + \overline{B})(A + \overline{C} + \overline{B})(\overline{A} + \overline{B} + C) \underline{(A + \overline{C} + C)} \\ &= (\overline{A} + \overline{B})(A + \overline{B} + \overline{C})(\overline{A} + \overline{B} + C) \quad \quad \quad = 1 \end{aligned}$$





Step 2: Find the missing variables in the common “OR-AND”, expand the terms to POM

$$\begin{aligned} F(A,B,C) &= (\bar{A} + \bar{B})(A + \bar{B} + \bar{C})(\bar{A} + \bar{B} + C) \\ &= (\bar{A} + \bar{B} + \bar{C})(\bar{A} + \bar{B} + C)(A + \bar{B} + \bar{C})(\bar{A} + \bar{B} + C) \\ &= (A + \bar{B} + \bar{C})(\bar{A} + \bar{B} + C)(\bar{A} + \bar{B} + \bar{C}) \end{aligned}$$

we can use the shorthand

$$F(A,B,C) = M_3 \cdot M_6 \cdot M_7 = \prod M(3,6,7)$$

Somehow, the algebra expression is of common “OR-AND”, we can proceed the second step directly.





II. Truth table conversion

1 . To obtain canonical SOM

The minterm expression of the logic function has a one-to-one correspondence with the truth table.

Assuming that the truth table of function F has k groups of variable combination whose values of F are 1, and the values of F of other combinations are 0; then the Minterms of function F is represented by “OR-ing” the value of the k group of variable combinations.

Method: In the truth table, find the variable terms whose values are 1, then, use “OR-ing” process to combine the terms. Thus, the SOM is obtained.





E.g. convert the logic function $F(A, B, C) = A\bar{B} + B\bar{C}$ to SOM

A: First, list the truth table of F;

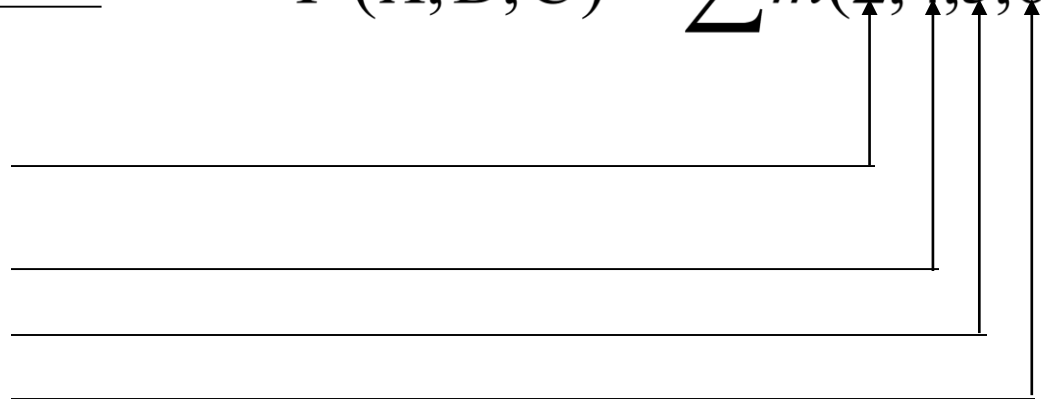
Then, use the truth table to obtain the SOM of F

$$F(A, B, C) = \sum m(2, 4, 5, 6)$$

Truth table

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

$$F(A, B, C) = \sum m(2, 4, 5, 6)$$





2 . To obtain canonical POM

The maxterm expression of the logic function also has a one-to-one correspondence with the truth table.

Assuming that the truth table of function F has k groups of variable combinations whose values of F are 0, and the value of F of other variable combinations are 1; then the maxterms of function F is represented by “ANDing” the value of the k group of variables.

Method: In the truth table, find the variable terms whose values are 0, then, use “ANDing” process to combine the terms. Thus, the POM is obtained.





E.g. Convert the logic function $F(A, B, C) = \overline{A}C + A\overline{B} \cdot \overline{C}$ to POM.

A: First, list the truth table of F;

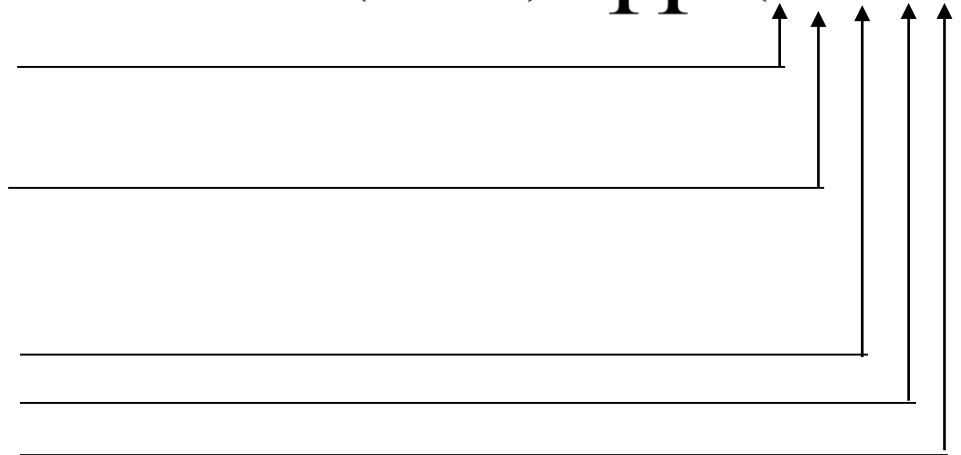
Then, use the truth table to obtain the POM of F

$$F(A, B, C) = \prod M(0, 2, 5, 6, 7)$$

Truth table of $F(A, B, C) = \overline{A}C + A\overline{B} \cdot \overline{C}$

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

$$F(A, B, C) = \prod M(0, 2, 5, 6, 7)$$





Function Complements

- The complement of a function expressed as a sum of minterms is constructed by selecting the minterms missing in the sum-of-minterms canonical forms.
- Alternatively, the complement of a function expressed by a Sum of Minterms form is simply the Product of Maxterms with the same indices.
- Example: Given $F(x, y, z) = \Sigma_m(1, 3, 5, 7)$
 $\bar{F}(x, y, z) = \Sigma_m(0, 2, 4, 6)$
 $\bar{F}(x, y, z) = \Pi_M(1, 3, 5, 7)$





Conversion Between Forms

- To convert between sum-of-minterms and product-of-maxterms form (or vice-versa) we follow these steps:
 - Find the function complement by swapping terms in the list with terms not in the list.
 - Change from products to sums, or vice versa.
- Example: Given F as before: $F(x, y, z) = \Sigma_m(1, 3, 5, 7)$
- Form the Complement: $\bar{F}(x, y, z) = \Sigma_m(0, 2, 4, 6)$
- Then use the other form with the same indices – this forms the complement again, giving the other form of the original function: $F(x, y, z) = \Pi_M(0, 2, 4, 6)$





Summary of Canonical Forms

Since there is a one-to-one correspondence between the function's truth table and the two canonical forms of the function; and the truth table of any the logic function is unique, **it can be seen that the two canonical forms of any the logic function are also unique.**

The uniqueness of the logic function expression brings great convenience to us to analyze and study the logic problem.

